

Beyond Masking: Landmark-based Representation Learning and Knowledge-Distillation for Audio-Visual Deepfake Detection

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Abstract

Audio-visual deepfake detection methods demonstrate strong performance on academic datasets but fail significantly when applied to real-world. To address the shortcomings of previous approaches, we utilize landmarks dynamic information. First, we propose Landmark-based Distillation (LBD), motivated by I-JEPA's representation learning approach. LBD utilizes KL-divergence to align facial landmark predictions from visual and audio encoders, enforcing focus on geometric facial features rather than spurious background information. Second, we introduce Multimodal Temporal Information Alignment (MTIA), which employs contrastive learning to enhance temporal consistency between audio and visual representations. We conduct experiments on academic datasets and web-based deepfakes collected from diverse social media platforms, serving as real-world examples. Our proposed landmark-guided distillation framework achieves computational efficiency while improving multimodal video deepfake detection performance across a diverse range of deepfakes compared to existing methods. The code is available at <https://github.com/Ckck12/Beyond-Masking>.

CCS Concepts

• Security and privacy → Social network security and privacy.

Keywords

Deepfake detection; Multimodal deepfakes; Real-world deepfakes

ACM Reference Format:

Chan Park, Muhammad Shahid Muneer, and Simon S. Woo. 2025. Beyond Masking: Landmark-based Representation Learning and Knowledge-Distillation for Audio-Visual Deepfake Detection. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*, November 10–14, 2025, Seoul, Republic of Korea. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3746252.3760853>

1 Introduction

The widespread and excessive use of AI-generated Content (AIGC) has fundamentally challenged the safety and privacy of current



Figure 1: Deepfake audio-visual samples were collected from social media platforms such as Meta, YouTube, X, and Reddit.

online ecosystems, as misinformation is prevalent using the latest AI-generated methods. Notably, recent studies [10] show significant increases in deepfake fraud attempts, with audio-visual manipulations [15, 33] becoming more sophisticated than unimodal approaches [18, 36–38]. We believe there will be more and more cases of creating fake news [26] motivated by political propaganda¹ and abuses [11] in the real world. In the recent Indo-Pak war, deepfake technology, specifically audio-visual deepfake² has been used to intensify conflict and build narratives for political agendas. While previous deepfake detection methods [1, 7–9, 14, 17, 21–23, 25, 29, 32, 33, 35, 41] demonstrate promising performance on popular benchmark datasets such as DFDC [5], FakeAVCeleb [16], and KoDF [20], they frequently exhibit substantial degradation and high false positives when facing real-world deepfake data [6, 14, 22, 23, 38]. Figure 3 shows a gradcam illustration, showing the backbone focusing more on spurious features, while the proposed framework, Landmark-based Distillation (LBD), focuses more on facial features. This implies that the existing methods learn sparse features, which the proposed method fills the gap. In this work, we identify two primary limitations in existing approaches. First, detection methods learn generation-specific patterns rather than exploiting the inherent consistency that characterizes real audio-visual content [30]. Second, previous approaches struggle with effective cross-modal alignment, failing to capture the fundamental relationships between facial movements and corresponding audio signals [30]. To address this, we leverage the key observation that real audio-visual video maintains predictable geometric and temporal relationships between facial features and audio signals.

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CIKM '25, Seoul, Republic of Korea

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ACM ISBN 979-8-4007-2040-6/2025/11

<https://doi.org/10.1145/3746252.3760853>

¹<https://www.youtube.com/shorts/DY6mPFIYKNI>

²<https://www.youtube.com/shorts/lr5rMeMG-P4>

Motivated by I-JEPA’s approach to learning semantic representations in embedding space [2], we introduce a novel landmark-guided self-distillation framework that aligns video and audio representations utilizing facial landmark representations as anchors. Our method comprises two components designed to effectively capture spatial and temporal consistency patterns of deepfakes. LBD addresses spatial consistency. We use facial landmarks as geometric anchors that guide both visual and audio encoders toward coherent representations through knowledge distillation. The intuition is that facial landmarks provide stable features that remain consistent in real content but exhibit discrepancies in deepfakes. Multimodal Temporal Information Alignment (MTIA) ensures temporal consistency across matching the audio-visual dynamics. We employ contrastive learning to align the temporal synchronization of audio-visual representations. MTIA captures the dynamic patterns that characterize real speech, where articulation and audio output must remain synchronized.

Our main contributions are summarized as follows:

- We propose the novel LBD module that guides encoders to focus on facial geometric features rather than spurious background information.
- We introduce MTIA using contrastive learning to enforce enhanced audio-visual temporal consistency.
- We demonstrate evaluation on five academic benchmarks and a newly collected dataset of real-world videos from the web, achieving SOTA performance over existing methods.

2 Methodology

2.1 Overview

Our landmark-guided self-distillation framework is depicted in Fig. 2. We extract video and audio features, respectively, while simultaneously obtaining facial landmark coordinates through MediaPipe. The landmark projector P_θ then transforms these coordinates into high-dimensional geometric representations. During LBD, landmark predictors P_ϕ^v and P_ϕ^a learn to align video and audio encoder outputs with the landmark-derived features through KL-divergence minimization. Subsequently, we apply transformer encoders with positional encoding to extract enhanced audio and visual features. For MTIA, we enforce temporal synchronization through contrastive learning between audio-visual pairs.

2.2 Problem Formulation

Let X be the input space of the videos $V \in \{V_j = 0, V_j = 1\}$, audios $A \in \{A_j = 0, A_j = 1\}$ and landmarks $L \in \{L_k = 0, L_k = 1\}$ consist of real and fake samples such that $y \in \{0, 1\}$. Given the source dataset, $\mathcal{D}_{src} = \{(x_i, y_j)\}_{i=1}^N$, the objective is to learn a 3D CNN f_{vid} , and a 1D convolution f_{aud} on $\mathcal{D}_{src}$ as follow:

$$\min_{f_{vid}, f_{aud}} \mathbb{E}_{(x, y) \sim \mathcal{D}_{src}} [\mathcal{L}_\phi^v(f_{vid}(x), P_\theta(L)) + \lambda_\alpha \mathcal{L}_\phi^a(f_{aud}(x), P_\theta(L)) + \lambda_\gamma \mathcal{L}_{MTIA}(A', V')], \quad (1)$$

where P_θ is a landmark predicted from the intermediate convolution attention of both audio and video, while $\mathcal{L}_\phi^v$ and $\mathcal{L}_\phi^a$ are the video and audio landmark reconstruction losses, respectively. The overall objective defined in Eq. 1 is to learn encoders f_{vid} and f_{aud} that

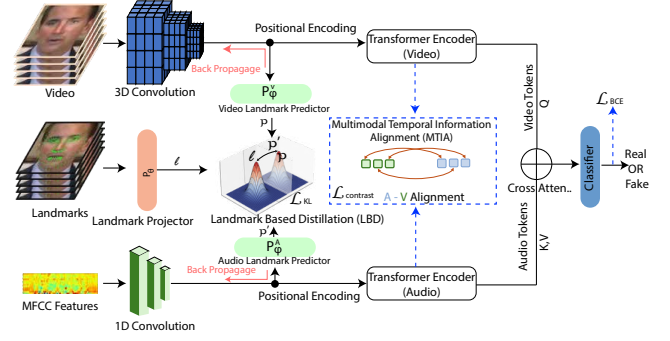


Figure 2: Overview of the landmark-guided framework with landmark predictors (P_ϕ^v , P_ϕ^a) and projector P_θ , and MTIA.

focus on deepfake-related artifacts of the target dataset $\mathcal{D}_{tgt}$ rather than sparse information in $\mathcal{D}_{src}$.

2.3 Landmark Based Distillation (LBD)

To achieve the objective of training models f_{vid} and f_{aud} we forward video V and audio A from the source dataset $\mathcal{D}_{src}$ in feed-forward functions as shown in Eq. 2 below to achieve intermediate attention A' and V' :

$$V' = f_{vid}(V), \quad A' = f_{aud}(A). \quad (2)$$

Similarly, we forward landmarks L to the landmark projector P_θ to obtain landmark projection l that can be adjusted to the same feature dimension as V' and A' from Eq. 2 for reconstructing using the landmark predictor P_ϕ^v and P_ϕ^a for video and audio, respectively, as follows:

$$p = P_\phi^v(V'), \quad p' = P_\phi^a(A'), \quad (3)$$

where V' and A' are the landmark approximation features that we further guide using Kullback-Leibler (KL) divergence loss for efficient reconstruction of landmarks embeddings following I-JEPA [3] and back propagation to enforce the model to learn to represent the features relevant to features in deepfake such as eyes, lips, and face attributes as shown below:

$$\mathcal{L}_{KL} = \text{KL}(\sigma(p/\mathcal{T}) \parallel \sigma(l/\mathcal{T})) + \lambda_\beta \text{KL}(\sigma(p'/\mathcal{T}) \parallel \sigma(l/\mathcal{T})), \quad (4)$$

where $\sigma(\cdot)$ denotes the row-wise softmax, $\mathcal{T}$ is a temperature hyperparameter, and λ_β is a loss weight that balances audio loss with the video loss.

2.4 Multimodal Temporal Information Alignment (MTIA)

As shown in Eq. 4, we trained the 3D Convolutional Neural Network (CNN) using landmarks L to focus on deepfake-relevant features rather than the sparse information, such as background, neck, or hair. In the following stages, we added positional encoding. We forwarded the features A' and V' extracted from Eq. 2 to the transformer encoder as shown in Eq 5:

$$\hat{V} = T(V' + PE), \quad \hat{A} = T(A' + PE), \quad (5)$$

Table 1: Intra-dataset evaluation results showing ACC, AUC, and model parameters on academic deepfake detection benchmarks. Bold and underlined indicate the best and second-best results, respectively.

Methods	Parameters	DF-TIMIT (LQ)		DF-TIMIT (HQ)		DFDC		FakeAVCeleb		KoDF	
		ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
SPSL (Xception) [24]	45.7 M	91.17	90.03	92.29	91.78	92.04	94.74	77.94	79.83	92.36	93.19
PRRNet [34]	62.1 M	93.86	<u>95.30</u>	<u>98.72</u>	98.97	95.24	<u>96.37</u>	78.17	79.54	94.52	94.83
LipForensics [13]	36.0 M	92.77	93.06	97.74	96.51	82.47	65.06	76.22	78.43	93.16	93.72
STIL [12]	32.5 M	<u>95.76</u>	94.18	98.24	99.03	92.17	95.07	79.80	78.29	<u>94.14</u>	93.92
FTCN [40]	26.6 M	93.64	94.41	96.62	95.39	<u>94.92</u>	97.88	78.98	79.21	93.22	92.96
DST-Net [39]	–	–	–	–	–	82.80	86.20	78.40	80.40	–	–
VFD [39]	–	–	–	–	–	80.90	85.10	81.50	<u>86.10</u>	–	–
Emotions [28]	42.7 M	92.76	92.95	94.69	95.42	82.36	85.00	<u>83.42</u>	84.81	93.26	94.41
MDS [9]	47.4 M	93.14	93.14	94.77	95.02	82.36	84.47	81.53	83.24	93.26	94.41
Joint Audio-Visual [41]	46.1 M	75.22	75.69	81.65	83.47	90.44	89.94	81.20	82.45	92.96	93.59
Ours	<u>32.4</u> M	96.82	97.35	98.98	<u>97.72</u>	89.75	90.07	92.38	93.25	93.48	<u>94.75</u>

where T and T' are video and audio transformer encoders, respectively, and PE is positional encoding for learning the temporal features of the video. Later, we forwarded $\hat{V}$ as Q and $\hat{A}$ as K and V to the multihead attention for joint learning and bringing audio and video features to exact joint feature representation V_A . The joint features V_A are then forwarded to a classification neural network head to classify as real or fake and compute binary cross entropy loss $\mathcal{L}_{BCE}$ for pushing real away from fake samples by backpropagating the weights to Eq. 2 for enforcing the model to learn the features that are different in real and fake on the landmark areas, rather than the whole image. To improve the generalization ability further, we trained contrastive objectives on $\hat{V}$ and $\hat{A}$ to further align the representation of both modalities as follows:

$$\mathcal{L}_{\text{contrast}} = \frac{1}{N} \sum_{i=1}^N \text{CE} \left(\frac{\hat{v}_i^T \hat{a}_j}{\|\hat{v}_i\| \|\hat{a}_j\|}, y_i \right), \quad (6)$$

where $\hat{v}_i$ and $\hat{a}_j$ represent normalized features from video $\hat{V}$ and audio $\hat{A}$, respectively, τ is a temperature parameter, CE denotes the cross-entropy loss, and $y_i = i$ is the index-based label for the positive pair in a batch of size N . This loss encourages matching embedding of audio-video pairs while pushing apart mismatched ones.

The following Eq. 7 is the final objective of the overall proposed representation learning framework that enforces the network to focus on learning robust representations learning of audio and visual deepfakes, as follows:

$$\mathcal{L} = \alpha(\mathcal{L}_{\text{KL}} + \mathcal{L}_{\text{contrast}}) + (1 - \alpha)\mathcal{L}_{\text{BCE}}, \quad (7)$$

where $\mathcal{L}$ is the final loss function, the overall objective of the representation learning framework. While α is a weighing factor that represents the weight that contributes during the training of the real or fake classification network.



Figure 3: Grad-CAM analysis comparing models trained with and without landmarks (green: real, red: fake).

3 Experimental Results

Benchmark Datasets. We followed existing audio-visual deepfake detection methods [4, 31, 32] on available datasets such as FakeAVCeleb [16], which contains 19,500 fake and 500 real videos. In contrast, fake samples contain either audio or video, but are either real or fake. The FakeAVCeleb dataset is generated using FaceSwap, FSGAN for facial forgery in videos, and SV2TTS for the generation of fake audio. To balance the training dataset with FakeAVCeleb, we utilized 10,000 VoxCeleb videos following [32]. Similarly, we used the KoDF[20], Deepfake-TIMIT [19], and the Deepfake Detection Challenge (DFDC) [5] for our experiments, which are publicly available and contain real and fake visual manipulations.

Real-World (RW) Web Dataset. We collected over 500 viral audio-visual deepfake samples sourced from major social media platforms, including Meta, YouTube, and Reddit. The criteria for choosing the audio-visual deepfake are the quality of the video and the realism of the audio. We also ensure that the audio-visual features of high-value professionals, such as presidents or military personnel, are evaluated to assess the performance of the approach in terms of robustness and relevant real-world high-impact scenarios. For real audio-visual samples, we used VoxCeleb 500 videos.

Experimentation Settings. First, we sampled 3 seconds of audio and video, containing 60 frames of video and 48 KHz of audio samples. Using the MediaPipe [27] library, we extracted 478 point

Table 2: Cross-dataset evaluation results. We test on DFDC and the proposed Real-World (RW) dataset. Bold and underlined indicate the best and second-best results, respectively.

Methods	Params	DFDC		RW Dataset	
		ACC	AUC	ACC	AUC
SPSL (Xception)	45.7 M	72.62	73.58	61.28	61.93
PRRNet	62.1 M	<u>74.65</u>	<u>75.18</u>	65.85	65.43
LipForensics	36.0 M	71.26	74.30	63.37	63.82
STIL	32.5 M	70.29	71.34	65.48	65.18
FTCN	26.6 M	73.18	74.03	64.19	64.80
Emotions	42.7 M	69.14	71.44	67.28	67.72
MDS	47.4 M	70.29	70.30	66.84	67.32
Joint Audio-Visual	46.1 M	71.22	72.69	<u>68.15</u>	<u>67.87</u>
Ours	<u>32.4 M</u>	78.12	78.82	75.12	75.65

landmarks on each X and Y axis and cropped faces from the video. To extract MFCC features, we used 40 channels, a hop length of 512, and an FFT length of 2,048. To enhance the understanding of the distinction between real and fake samples, we paired real samples with fake samples from the same manipulated videos. We kept the value of $\alpha = 0.8$, $\lambda_\alpha = 0.5$, $\lambda_\gamma = 0.5$, and $\lambda_\beta = 0.5$.

3.1 Intra-dataset Performance

We evaluate intra-dataset performance using 80% and 20% for train and test data splits across five benchmark datasets. We compare against representative baselines from two categories: visual unimodal methods [12, 13, 24, 34, 40], and multimodal approaches [9, 28, 39, 41]. As shown in Table 1, our method achieves competitive performance across all datasets with fewer parameters. We particularly excel on FakeAVCeleb, which involves manipulations in both modalities simultaneously or solely, as our LBD and MTIA components effectively capture multimodal inconsistencies that unimodal methods miss. This demonstrates strong performance on complex audio-visual manipulations.

3.2 Evaluation on Unseen Cross-dataset

Table 2 shows cross-dataset evaluation results to assess generalization capabilities across unseen data. We train on the FakeAVCeleb and KoDF and evaluate on the DFDC and real-world datasets that we have collected. While baseline methods achieve competitive performance in intra-dataset evaluation on Table 1, they suffer significant performance degradation in cross-dataset DFDC and real-world scenarios. This illustrates the generalization challenge that existing approaches encounter when faced with unseen data distributions. Notably, unimodal methods show performance degradation due to their inability to detect audio-visual inconsistencies that are prevalent in real-world deepfakes. In contrast, our proposed framework demonstrates superior cross-dataset performance. Against multimodal baselines, we achieve better generalization with fewer parameters by focusing on geometric features rather than spurious patterns through landmark-guided attention. These results validate that combining geometric consistency with self-supervised learning enhances generalization across different data distributions.

Table 3: Ablation study on effects of different combinations of P_ϕ^V , P_ϕ^A , P_θ , and MTIA.

P_ϕ^V	P_ϕ^A	P_θ	MTIA	FakeAVCeleb	
				ACC	AUC
$\times$	$\times$	$\times$	$\times$	72.12	71.81
$\checkmark$	$\times$	$\checkmark$	$\times$	82.08	81.57
$\times$	$\checkmark$	$\checkmark$	$\times$	75.89	75.14
$\checkmark$	$\checkmark$	$\times$	$\times$	84.25	85.60
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	89.43	90.15
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	92.38	93.25

3.3 Ablation Study

Table 3 shows the individual contribution of the combination components through ablation analysis: P_ϕ^V , P_ϕ^A , P_θ , and MTIA. The baseline (row 1) utilizes each modality encoder separately and concatenates the features through a fully connected layer. Comparing individual landmark predictor combinations (rows 2-3), the visual $P_\phi^V + P_\theta$ outperforms the audio counterpart $P_\phi^A + P_\theta$. This performance gap stems from the reduced domain heterogeneity between visual and landmark modalities compared to audio-landmark alignment, making visual-landmark distillation more effective. The combination of both landmark predictors without projection (row 4) demonstrates the benefits of self-distillation, while adding landmark projection (row 5) further improves performance through enhanced feature alignment. Leveraging all components, it is crucial to validate that temporal synchronization alignment between visual and audio dynamics is accurate for detecting deepfake inconsistencies. To further visualize the effectiveness of the proposed approach, we draw GradCAM in Fig. 3, in which we can see how the LBD helps in focusing on facial features rather than background.

4 Conclusion

Our approach demonstrates its effectiveness in mitigating the sparse behavior of encoders by using self-distillation conditioned on landmarks, which helps the encoder focus on more deepfake-related features rather than background or other irrelevant parts of the image. We demonstrate through experiments that the our method effectively reduces the sparse focus of encoders on irrelevant features. By utilizing landmarks for self-distillation, we establish a relationship within the encoders to focus more on facial features, thereby improving the synchronization. This effectively enhances the generalization ability of the proposed approach, enabling the detection of real-world audio-video deepfakes.

Acknowledgments

This work was supported by Institute for Information & communication Technology Planning & evaluation (IITP) grants funded by the Korean government MSIT: (RS-2022-II221199, RS-2022-II220688, RS-2019-II190421, RS-2023-00230337, RS-2024-00356293, RS-2024-00437849, RS-2021-II212068, RS-2025-02304983, RS-2025-02263841, RS-2024-00436936, RS-2024-00437849).

GenAI Usage Disclosure

We used GenAI tools to assist with equation corrections and GPT-4o for PyTorch boilerplate code. AI usage was limited to improving presentation clarity and grammar check.

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